

Spinal Cord Syndromes and Common Mononeuropathies

Adena Leder, DO, FAAN

Associate Professor

Department of OMM

aleder@nyit.edu

Do.
Make.
Heal.

Innovate.

Reinvent the Future.

Session Objectives

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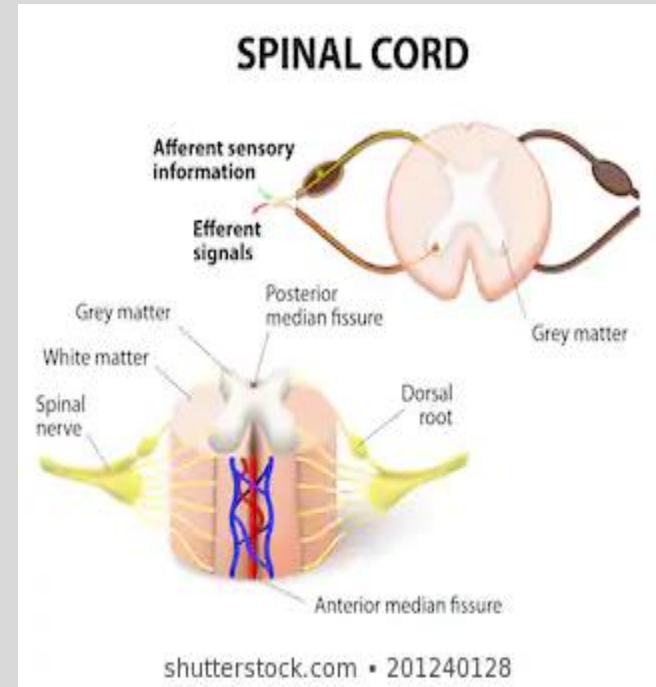
- Objective 1-**Describe** the ascending and descending spinal cord tracts.
- Objective 2-**Explain** common spinal cord syndromes based on cord location.
- Objective 3- **Generate** a differential diagnosis based on the location of the spinal cord injury.
- Objective 4- **Apply** the proper work up for a spinal cord lesion.
- Objective 5- **Recognize** a neurosurgical emergency.

A Review of Cord Anatomy

- Cord ends at L1
- Central grey matter-
 - Contains neuronal cell bodies and synapses.
- Peripheral white matter.
 - Contains ascending and descending fiber pathways.

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Ascending Tracts

- Posterior Columns
 - **Gracile** fasciculus-**lower** body
 - **Cuneate** fasciculus-**upper** body
 - Mediates fine touch, vibration and **proprioception- crosses over in the brainstem.**

Where am I in space?

- Spinothalamic tracts-crude touch/pain/temp-cross over after ascending one or 2 levels.

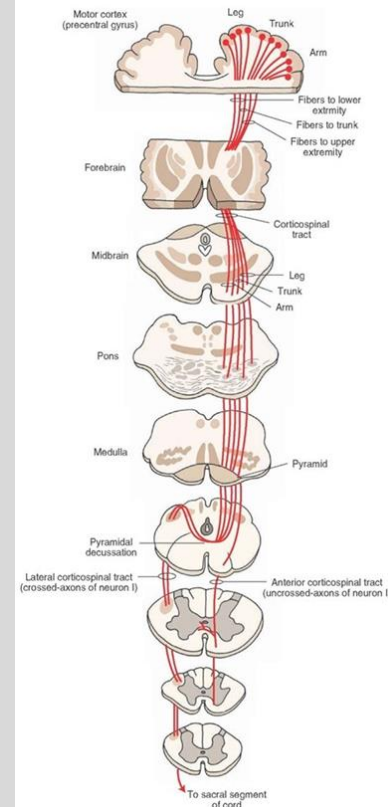
Descending Tract

- **Corticospinal tract**- begins in primary motor cortex.
- Long tract- primary motor cortex in the precentral gyrus to the anterior horn.
- At the level of the medulla the majority cross over and become the lateral corticospinal tracts.

Localization in Clinical Neurology, Wolters Kluwer, eighth edition, 2022

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Myelopathy-signs/symptoms

Common signs and symptoms:

- Gait disturbance
- Spasticity
- Weakness
- Hyperreflexia
- Bladder or bowel symptoms
- Sensory Level- a bandlike sensation at the superior margin of the level. There is numbness and paresthesia below that level.

Types of Spinal Cord Lesions

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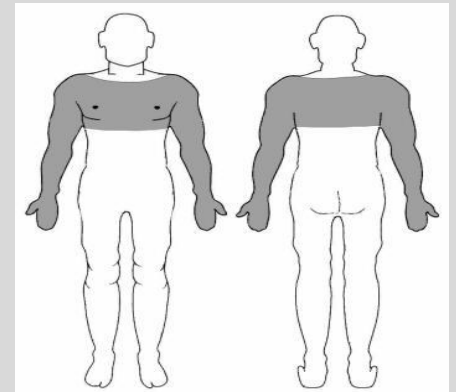
- Central Lesion (syringomyelia)
- Posterior Column Syndrome (tabes dorsalis)
- Anterior Cord Syndrome (ant.spinal art)
- Cauda Equina Syndrome
- Posterolateral Column Syndrome (Subacute Combined Degeneration)
- Complete transection
- Brown Sequard Syndrome

Etiologies

- Vascular
- Infectious
- Traumatic
- Autoimmune
- Metabolic/Toxic/Nutritional
- Inflammatory
- Neoplastic
- Structural

Case 1

- A 30-year-old otherwise healthy woman presents to an outpatient clinic complaining of **loss of pain and temperature in a capelike distribution.**
- Sensory and motor are also affected in the hands.



Central Cord Syndrome

- Affects the gray matter in the center of the cord
- Causes:
 - Trauma (severe cervical hyperextension)
 - Intramedullary tumor
 - **Syrinx (may also have Chiari)**
 - Small hemorrhages
 - Demyelinating lesion

Central Cord/Syrinx

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- A CSF containing cord cavity.
- Cervical region most commonly affected.
- Rarely in thoracic cord alone.
- Seen in 20-40% of Chiari I malformations.

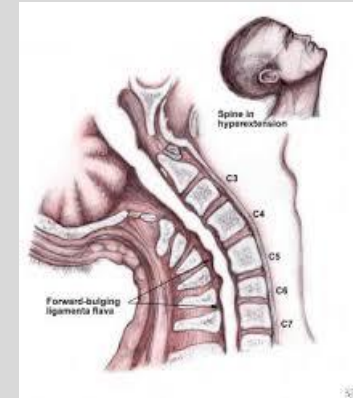
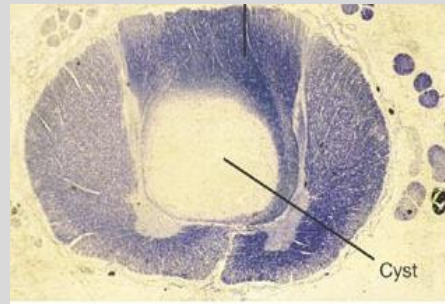
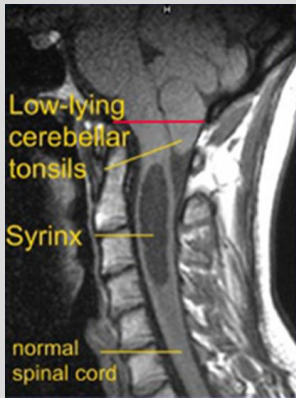


Image- emedicine.Medscape Localization in Clinical
Neurology. Wolters Kluwer. eighth edition. 2022

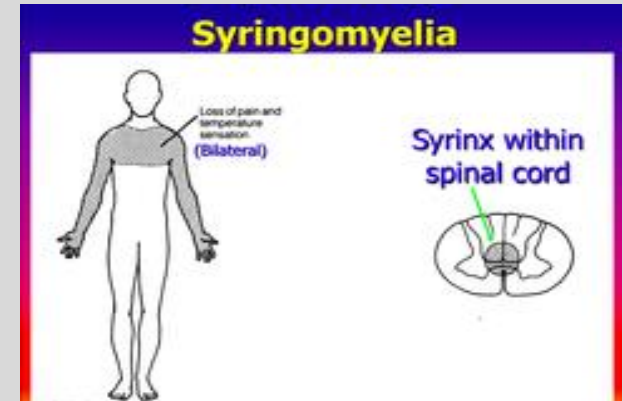
Syringomyelia

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- Cord damage begins centrally and spreads outward.
- Spinothalamic fibers damaged initially.
- Loss of pain and temp in capelike distribution.

<https://quizlet.com/19801505/head-injury-and-spinal-trauma-flash-cards> American Academy of Neurology-Continuum series- Spinal Cord Disorders, February, 2024 VOL 30, No 1



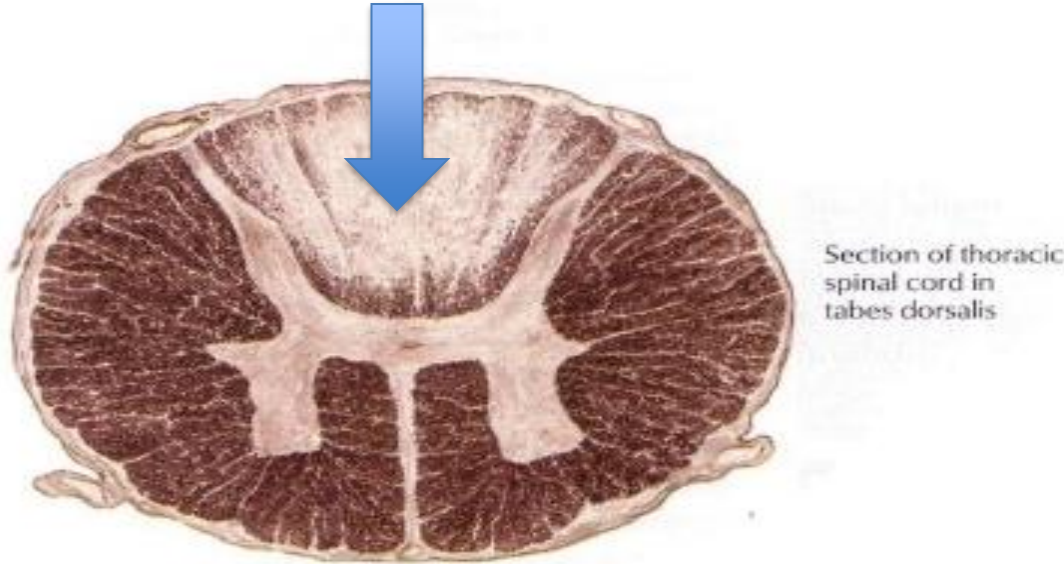
Case 2

- An 80-year-old man presents with difficulty walking.
- He states that he has fallen while in the shower (especially when he closes his eyes to wash his hair).

Dorsal Column Disorder

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http://library.med.utah.edu/neurologicexam/html/gait_abnormal.html#14

Dorsal Column Syndrome

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- Impairment of vibration and proprioceptive sense.
- “Stomping gait”
- Lhermitte sign-electric shock running down the back-elicited by neck flexion.
- Dorsal root involvement → absent reflexes knees and ankles with preserved strength.
- Loss of joint sensation/ repeated microtrauma → Charcot arthropathy

Etiology of Dorsal Column Syndrome

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- Tabes Dorsalis-syphilitic spinal cord disease
- Cervical spondylosis
- Posterior spinal artery infarction
- Multiple Sclerosis
- Partial cord trauma
- Post platinum-based chemo



Image: Ooscskills.com/ American Academy of Neurology- American Academy of Neurology- Continuum Spinal Cord Disorders, February, 2024 VOL 30, No 1

Tabes Dorsalis-Neurosyphilis

- Caused by the spirochete *T. pallidum*.
- Occurs decades after primary infection (most common after 20 years).
- Progressive gait abnormalities-slapping
- Positive Romberg- swaying with the eyes closed

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Image: Oskill.com American Academy of Neurology- Continuum series- Spinal Cord Disorders, February, 2024 VOL 30, No 1

Symptoms of Tabes Dorsalis

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- Triad of symptoms:
 - Lightning or (lancinating) pains
 - Ataxia
 - Bowel and bladder dysfunction
 - Triad of signs:
 - Argyll Robertson pupils- small pupils that react to accommodation, but not to light
 - Areflexia
 - Loss of proprioception.

Lhermitte Sign



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Image: Youtube.com

Case 3

- A 58-year-old woman underwent bariatric surgery several months ago. She was non-compliant in following up with her surgeon.
- Today she presents with numbness and weakness of her feet.

Posterolateral Column

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- Involvement of posterior column and lateral corticospinal tracts.



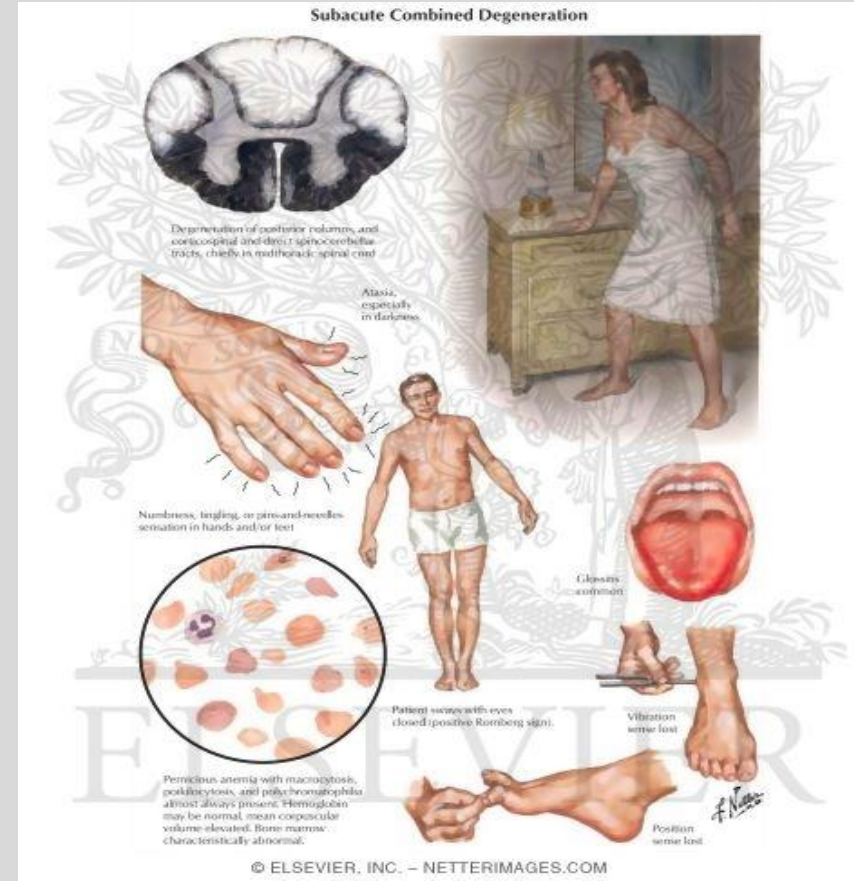
Netter plate

Subacute combined degen

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- Numbness
- Positive Romberg sign
- Glossitis
- Pernicious anemia
- Vibration and position sense lost

Netter plates- subacute combined degeneration



Causes of SCD

- Vitamin B12 deficiency
- Copper Deficiency
- Cervical spondylosis
- HTLV-1 associated myelopathy/tropical spastic paraparesis
- AIDS-associated myelopathy
- Nitrous oxide exposure
- Heroin use (IV or by inhalation)

Causes of B12 Deficiency

- Bariatric surgery
- Atrophic gastritis
- Celiac disease
- Pancreatic disease
- Vegetarian diet
- Antacid use
- Nitrous oxide toxicity or abuse
- Infections-H. Pylori, giardia, HIV, D. latum

Nitrous Oxide

- Laughing gas, colorless odorless
- Causes irreversible inactivation of B12
- Recreationally used for mood elevating properties.
- Toxicity-after repeated use or after a single exposure.
- May cause a myelopathy, cognitive issues myeloneuropathy, peripheral neuropathy.
- Posterior and lateral columns of cervical and thoracic cord are affected.

Case 4

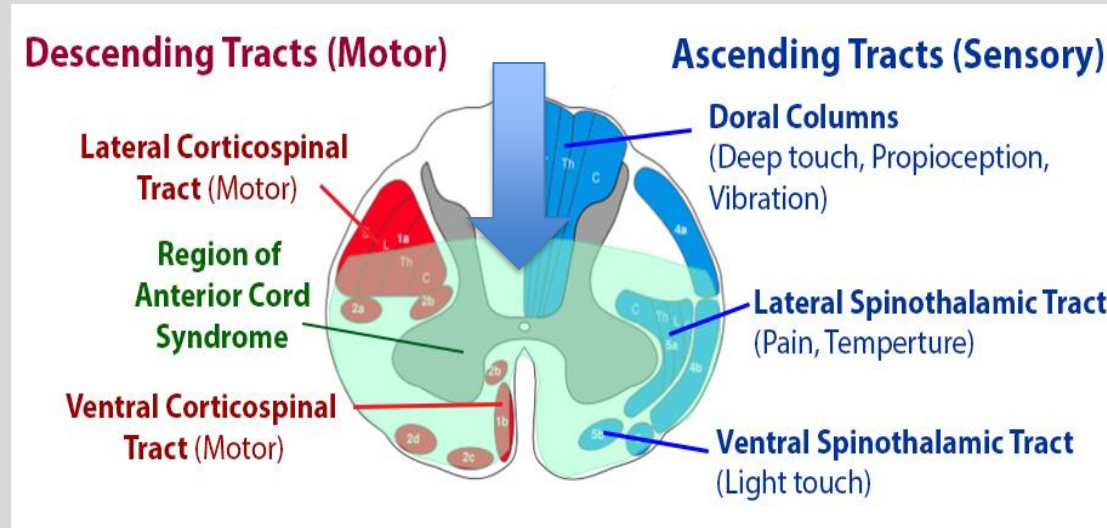
- 73-year-old man
- Abdominal aortic aneurysm rupture
- Surgical repair of aorta
- Next morning *could not move legs*
 - *Flaccid paraplegia*
 - *Sensory level T9-T10*
 - *Anesthesia (numbness to pain and temp below that level)*

Anterior Cord Syndrome

- Weakness and numbness
- Proprioception is spared

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Question

- What is the most common cause of a spinal cord infarction?
 - a) Atherosclerosis of aorta
 - b) Car accident
 - c) Aortic surgery
 - d) Embolization

Ans A

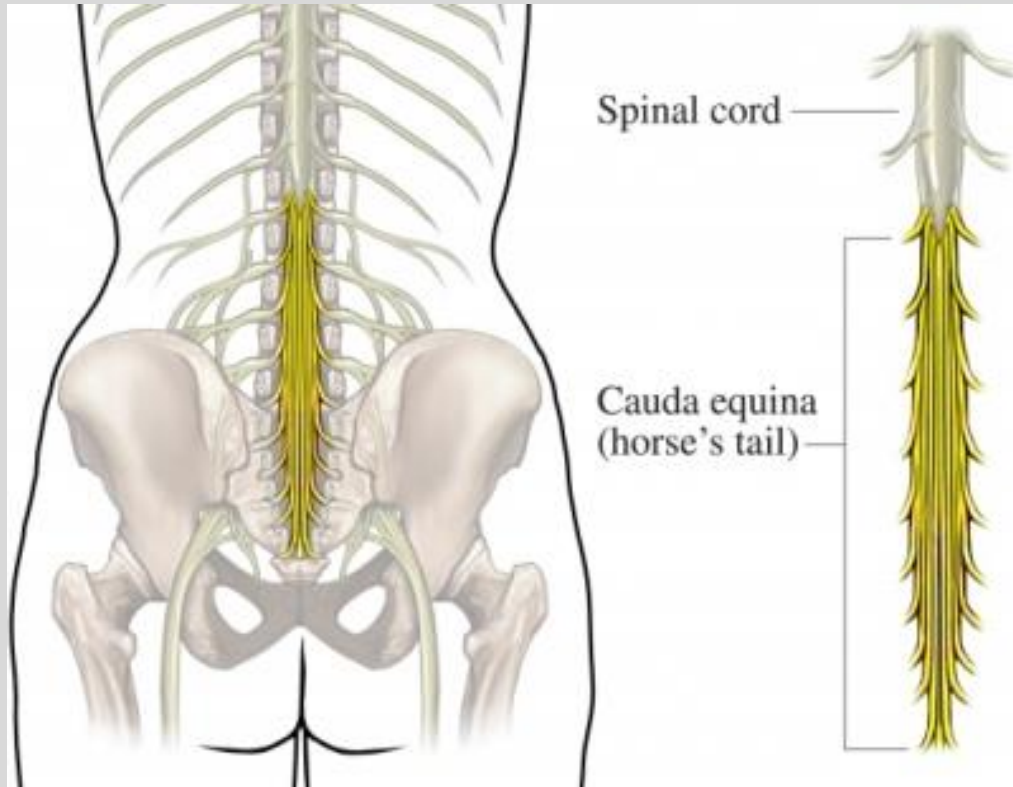
Clinical pattern Ant. Cord

- Weakness below the level of the lesion
- Loss of pain and temperature sensation below level of the lesion
- Autonomic function loss below level.
- Vibration and position sensation spared. Most suggestive of an anterior spinal artery infarct.

Case 5

- A 68-year-old man presents to the ED with severe low back pain and radiation into bilateral posterolateral legs (right >left).
- He admits to weakness in the legs.
- Difficulty with urination for the past few days.
- Cannot feel when he is urinating.

Cauda Equina



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Premier surgery center of Louisville-cauda equina syndrome

Cauda Equina

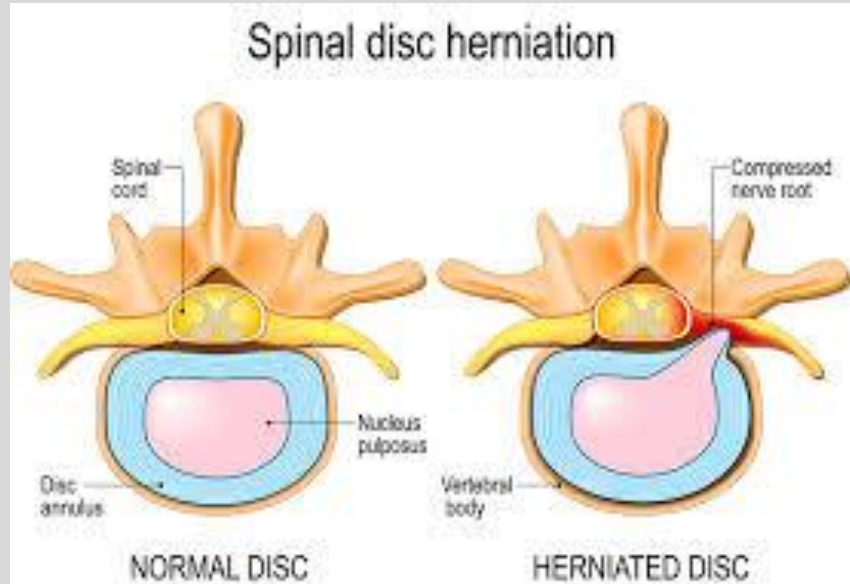
- Collection of nerve roots-
- L2-L5
- S1-S5
- Coccygeal segments
 - Lower limb movement- L2-S2
 - Lower limb sensation- L2-S3
 - Bladder control- S2-S4
 - Control of external anal sphincter S2-S4
 - Perineal sensation- S2-S4

Causes of Cauda Equina

- Intervertebral disc herniation (MC)
- Spinal fracture
- Iatrogenic cause
- Tumor

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Clinical Features CE

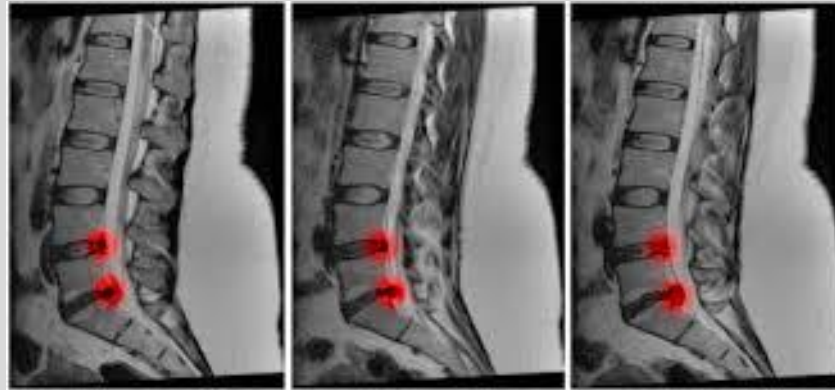
- Low back pain and radicular symptoms
- Bladder symptoms
- Bowel symptoms (less than urinary symptoms- constipation, rectal fullness)
- Sexual dysfunction
- Sensory loss- buttocks, posterior thighs, perineal region- MUST ASK!
- Weakness- **asymmetric**
- Reflexes- Reduced at knees and ankles.

Assess and Manage CE

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- MRI is the imaging modality of choice.
- Treatment is surgical decompression.
- Only occurs in 1-2% of cases.



Case 6

- A 49-year-old man presents to the ED after a bar fight. It resulted in a knife to his spine on the right.
- On examination there is weakness on the right side with loss of pain and temperature on the left side below the level of the lesion.

Brown Sequard Syndrome

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- Hemicord injury
- Disrupts descending ***already crossed*** corticospinal fibers
 - Segmental muscle weakness and atrophy and sensory signs.
 - Ipsilateral motor weakness
 - Ipsilateral vibratory/proprioceptive loss
 - Contralateral pain/temperature below the lesion.
- ***Often incomplete***

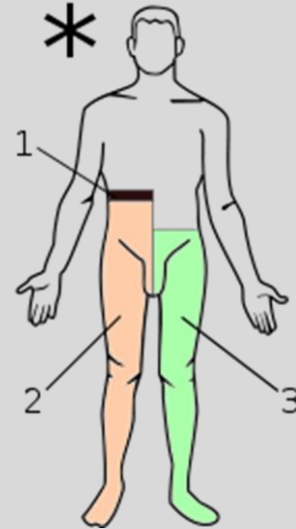


Image: quora.com Localization in Clinical Neurology, Wolters Kluwer, eighth edition, 2022

Check in...

- In a hemicord lesion (Brown-Sequard) what type of lesion is most likely responsible?
- a) Intramedullary
- b) Extramedullary
- c) Hematogenous spread

Ans.B

Causes of Hemicord Syndrome

- Penetrating trauma/stab wound
- Multiple sclerosis
- Spinal subdural hematoma
- Asymmetric compression
- Neurosarcoidosis/other inflammatory infectious
- Neoplasm (primary or metastatic)
- Spinal cord herniation

Case 7

- A 16-year-old man was brought to the hospital after doing a trick on his bicycle.
- When he landed, he fractured his cervical spine at the level of C7 (below the phrenic nerve).
- The spinal cord was crushed. As a result, he developed paralysis below C7, loss of sensation and loss of autonomics (could not have a bowel movement without enema and disimpaction).

Complete Transection

- Trauma (stab wound)
- Demyelinating Disease
- Spinal cord vascular disorders
- Epidural abscess or metastasis
- Paraneoplastic necrotizing myelopathy
- Late radiation-induced myelopathy
- Herniated intervertebral disc

Complete Transection

- Bilateral upper motor neuron pattern weakness below level
- Bilateral sensation loss below level
- Autonomic function loss below level (constipation most common).

Infectious Myelopathies

- Microorganisms can also cause dysfunction to the spinal cord.
 - Viral
 - Bacteria
 - Parasites
 - Fungal infections

Viruses in Myelopathy

- HIV
- Herpesviruses
- Enteroviruses
- HTLV
- Varicella-zoster virus
- Cytomegalovirus
- Epstein Barr virus
- West Nile virus
- Polio virus
- Dengue virus
- St. Louis and Japanese Encephalitis

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Covid-19 and Myelitis

- **Acute myelitis as a neurological complication of Covid-19: A case report and MRI findings**
- [Reem AlKetbi](#), MD,^{a,*} [Dana AlNuaimi](#), MD,^a [Muna AlMulla](#), MD,^a [Nouf AlTalai](#), MD,^b [Mohammed Samir](#), MD,^a [Navin Kumar](#), MD,^a and [Usama AlBastaki](#), MD^a

Review of cord lesions

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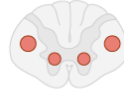
Spinal cord lesions

AREA AFFECTED

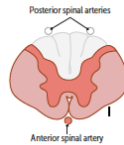
DISEASE



Spinal muscular atrophy



Amyotrophic lateral sclerosis



Complete occlusion of anterior
spinal artery



Tabes dorsalis



Syringomyelia



Vitamin B₁₂ deficiency

Common Mononeuropathies

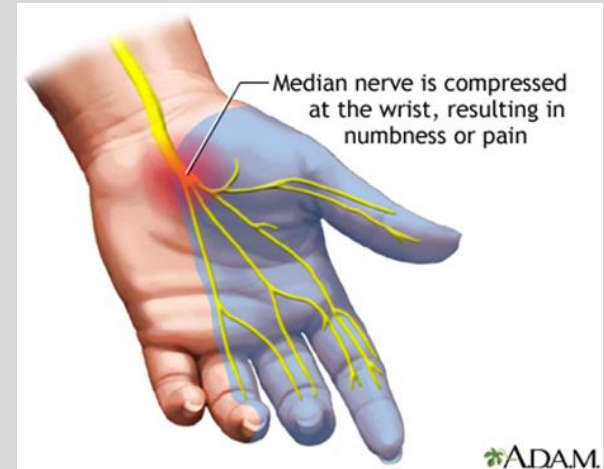
- Median Neuropathy
- Ulnar Neuropathy
- Radial Neuropathy
- Fibular (peroneal)
- Lateral femoral cutaneous nerve

Median Neuropathy

- Carpal tunnel syndrome
- Wrist pain
- “Hand numbness”-sensory loss in the palmar aspect of the first 3 or 4 digits.
- Thenar weakness late in the course of the condition.

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Ulnar neuropathy

- Numbness in ½ 4th and 5th digit
- Clawing of 4th and 5th digit
- Weakness and wasting of the first dorsal interosseous muscle
- Repeated elbow trauma
- Finger Adductor weakness

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Radial Neuropathy

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- Unilateral painless wrist drop
- Finger or wrist extension weakness
- Painless gradually progressive bilateral wrist drop- lead poisoning.
- Alcoholic with Saturday night palsy (spiral groove is localization)

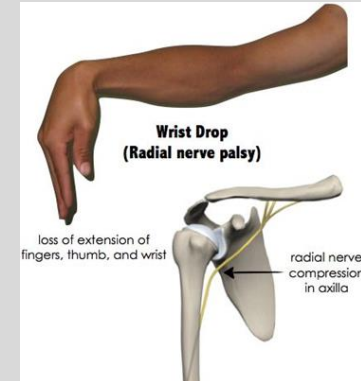
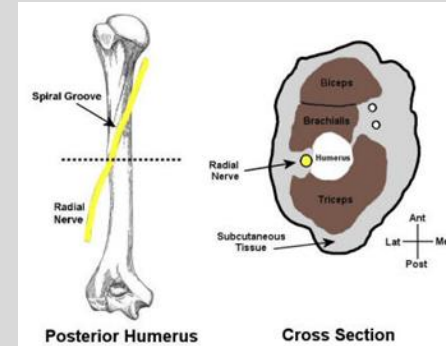
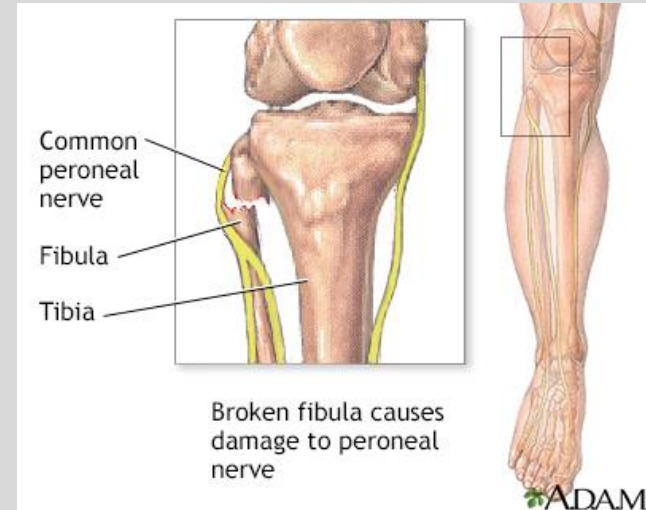


Image:school of medicine- case western/pinterest
Instant neurological diagnosis- Hawkes, Sethi, Swift

Fibular (peroneal neuropathy)

- Most frequent mononeuropathy in the lower extremity.
- Weakness of ankle dorsiflexion with foot drop may be seen.
- Trauma is the most common cause at the fibular head.
- May be seen with recent weight loss.



Meralgia Paresthetica

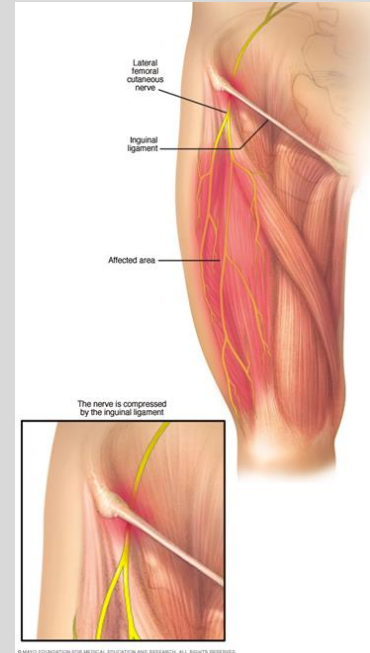
- Compression of the lateral femoral cutaneous nerve where it crosses under the inguinal ligament.
- Numbness of the lateral thigh.
- Pregnancy, weight gain, pelvic surgery

Image: mayo clinic

Instant neurological diagnosis-Hawkwa, Sethi, Swift

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Summary Slide

- Recognize spinal cord syndromes based on clinical presentation.
- Apply proper work-up and recognize neurosurgical emergencies.
- Core References:
 - American Academy of Neurology- Continuum series- Spinal Cord Disorders, February, 2024 VOL 30, No 1
 - Localization in Clinical Neurology, Wolters Kluwer, eighth edition, 2022

Lecture and Lab Feedback Form:

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<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>